

Lab 8: Security and Firewall Management

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This article details the commands used to improve the security and firewall of the Ubuntu Server. This article walks through all of the commands used, their purpose, and includes screenshots detailing the correct results of the commands that were run. In the first part it goes through Security Checks and Configurations, while the second part goes through the Firewall Checks and Configurations.

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VM Snapshot(s)

Virtual Machine 1542 (CIT371-001-Fall2025-draperj2-Ubuntu22Server) on node 'COIVMHOST2' cit371-001-fall2025 ▶ Start

Summary

Console

Hardware

Cloud-Init

Options

Task History

Snapshots

Firewall ▶

Take Snapshot | Rollback | Edit | Remove

Name	RAM	Date/Status	Description
🔄 Day1	No	2025-08-07 10:07:19	Used to revert VM to initial state. Do not delete!
🔄 Lab2	No	2025-09-04 12:44:29	Lab2
🔄 Lab3	No	2025-09-12 17:34:53	Lab3
🔄 Lab4	No	2025-09-13 11:00:54	Lab 4
🔄 Lab6	No	2025-10-01 13:53:21	Lab 6
🔄 Lab7	No	2025-10-08 13:48:03	Lab 7
🔄 Lab8	No	2025-10-09 17:22:43	Lab 8
🖥️ NOW			You are here!

Overview

In this lab I performed several key security and firewall configurations to help secure the Ubuntu Server. I first performed a system update and upgrade. This is important to ensure all packages are current and secure. This process installs the latest security patches, fixes vulnerabilities, and keeps the system protected. I then created a test user without a password and identified the accounts lacking passwords. This is important because password-less accounts can pose a major security risk by allowing anyone to log in. After that I deleted the account, doing this helps maintain proper access control and prevents potential exploits. I then created and encrypted a file using GPG to secure sensitive data. Encryption can help make sure that even if a file is accessed, the contents are unreadable without the passphrase. I then disabled password caching for the GPG agent, that way the passphrases would not be stored in memory. Of course, this increases security by requiring the user to enter the correct passphrase each time they try to access the file. Next, I edited the SSH configuration file to disable root login. This can reduce the server's attack surface by making users log in with their own accounts before they can perform any administrative tasks.

Following that I enabled the UFW to control incoming and outgoing traffic. Enabling the firewall provides a strong defense against network-based attacks. I also viewed the active firewall rules using the verbose option and enabled UFW logging. Looking at current rules helps you to confirm the correct configurations. I then denied access through port 23, in order to disable Telnet connections. Since Telnet only sends data in plain text, not encrypted text, blocking it is essential to prevent data breaches. Next, I created a numbered rule to allow traffic on port 8080. This allows web traffic for testing or development environments. Lastly, I added firewall rules that allow only my client system to access ports 20 and 21. This limits FTP access and enhances security by making sure only authorized connections are approved.

Procedure

The first step in this lab is to sign in to the Proxmox cluster and access the Ubuntu Server; then open a terminal.

Part 2: Security Checks and Configurations

The first command you will use updates and upgrades the server:

```
sudo apt update
```

```
sudo apt upgrade -y
```

Then enter the following commands to create a password-less user:

```
sudo useradd testuser
```

```
sudo passwd -d -u testuser
```

Then enter the following command to find the user created without a password:

```
sudo awk -F: '($2==""){print $1}' /etc/shadow
```

The output should look like this:

```
student@371serv:~$ sudo awk -F: '($2==""){print $1}' /etc/shadow
testuser
student@371serv:~$
```

Next, we will delete the password-less account with this command:

```
sudo userdel testuser
```

Then, switch to the root user with this command:

```
sudo -i
```

Next, we are going to create a file using this command:

```
echo "Secure Server" > /root/secure.txt
```

Then we are going to encrypt the file with GPG using this command:

```
gpg -c /root/secure.txt
```

When prompted for a passphrase use this: *Passw0rd*

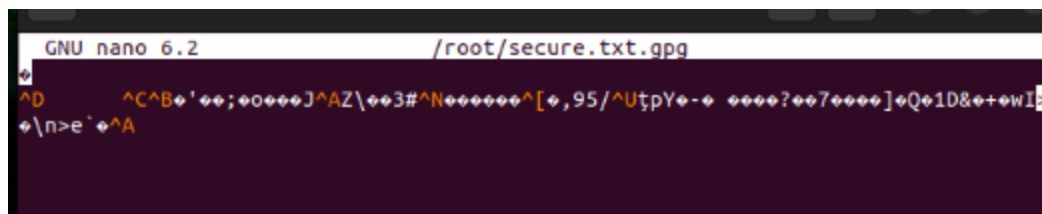
We will then remove the unencrypted version of the file with this command:

```
rm /root/secure.txt
```

Then open the file in nano to ensure it was encrypted using this command:

```
nano /root/secure.txt.gpg
```

The file should look like this:



Then sign out from the root user with this command:

```
exit
```

Then we want to disable password caching for the GPG agent in the student user's home directory with these commands:

```
echo "default-cache-ttl 0" > ~/.gnupg/gpg-agent.conf
```

```
echo "max-cache-ttl 0" >> ~/.gnupg/gpg-agent.conf
```

Finally, we want to disable root login for ssh with this command:

```
sudo nano /etc/ssh/sshd_config
```

In the nano file, find the line 'PermitRootLogin'. Remove the comment from this line and write 'no' after 'PermitRootLogin'.

Save the file by pressing Ctrl+O, Enter, Ctrl+X.

Reset the sshd service using this command:

```
sudo systemctl restart sshd
```

Part 3: Firewall Checks and Configurations

Run this command to enable the UFW firewall:

```
sudo ufw enable
```

The output should look like this:

```
student@371serv:~$ sudo ufw enable
Firewall is active and enabled on system startup
student@371serv:~$
```

Then run this command to list the firewall's current list of configured ports using the verbose option:

```
sudo ufw status verbose
```

The output should look like this:

```
student@371serv:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)

student@371serv:~$
```

Next, enable UFW logging using this command:

```
sudo ufw logging on
```

The output should look like this:

```
student@371serv:~$ sudo ufw logging on
Logging enabled
student@371serv:~$
```

We will then deny access through port 23 to block telnet connections using this command:

```
sudo ufw deny 23
```

The output should look like this:

```
student@371serv:~$ sudo ufw deny 23
Rule added
Rule added (v6)
student@371serv:~$
```

Then we want to create a numbered rule that allows port 8080 using this command:

```
sudo ufw insert 1 allow 8080
```

The output should look like this:

```
student@371serv:~$ sudo ufw insert 1 allow 8080
Rule inserted
Rule inserted (v6)
student@371serv:~$
```

Next, we want to write two rules that only allow our client to access TCP ports 20 and 21 (replace the IP address with your IP address):

```
sudo ufw allow from 192.168.1.5 to any port 20 proto tcp
```

```
sudo ufw allow from 192.168.1.5 to any port 21 proto tcp
```

The output should look like this:

```
student@371serv:~$ sudo ufw allow from 192.168.1.5 to any port 20 proto tcp
Rule added
student@371serv:~$ sudo ufw allow from 192.168.1.5 to any port 21 proto tcp
Rule added
student@371serv:~$
```

This completes the lab; you have finished successfully. Exit the terminal and sign out of the Proxmox cluster.

References

DigitalOcean:

[How to Set Up a Firewall with UFW on Ubuntu | DigitalOcean](#) 

nixCraft:

[How to List UFW firewall rules on Linux - nixCraft](#) 

Unix & Linux Stack Exchange:

[sudo - User without a password - how can one login into that account from a non-root account - Unix & Linux Stack Exchange](#) 

Ubuntu:

[Firewall - Ubuntu Server documentation](#) 

